MIL-P-50265
21 April 1969

MILITARY SPECIFICATION

PRIMER, ELECTRIC, XM114
PARTS FOR, AND LOADING, ASSEMBLING, AND PACKING

1. SCOPE

1.1 This specification covers the inspection of the parts and the loading, assembling and packing of one type of electric initiating device designated as Primer, Electric, XM114.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U. S. ARMY MUNITIONS COMMAND

8796522 - Marking Diagram and Sealing for Wood Packing Boxes.

FSC: 1375

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9225836 - Primer, Electric, XM114 Assembly.

PUBLICATIONS

U. S. ARMY MUNITIONS COMMAND

ET-9225836 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

Code of Federal Regulations, Title 49-Transportation, Parts 0-190

(The Code of Federal Regulations is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402. Orders shall specify 49 CFR 0-190. (latest revision).)

3. REQUIREMENTS

3.1 Material and parts.- Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Base charge moisture content.- The moisture content of the base charge at the loading station at the time of loading shall not exceed 0.5 percent when tested as specified in 4.4.1.

3.3 Functioning.- The primer assemblies shall not fail to function on the first attempt and shall comply with the following:

3.3.1 Time.- The primer assemblies shall fire within three hundred (300) microseconds when tested as specified in 4.4.2.1.

3.3.2 Sound power output.- The sound power output shall be a minimum (min.) of one and four tenths (1.4) pounds per square inch (p.s.i) when tested as specified in 4.4.2.2.

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3.3.3 Primer assembly fails to fire on first attempt (Dud).- The primer assembly shall not dud when tested as specified in 4.4.2.3.

3.4 Freon TF and water immersion.- After Freon TF and water immersion, the primer assembly shall comply with the pin and cup resistance requirements of dwg. 8225836 and the sound power output requirement of 3.3.2. The test shall be performed as specified in 4.4.6 and then in accordance with the firing requirements of 4.4.2.2.

3.5 Workmanship.- All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, surface defects, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

3.6 First article inspection.- This specification makes provisions for first article inspection. Requirements for the submission of first article samples shall be specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.- At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

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b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specification referenced in any of the contractual documents, and that certificates of conformance are on file and available for review, except that a statement of findings on the moisture content of the base charge shall be furnished with each lot submitted for acceptance.

d. Number of items in the lot.

e. Date submitted

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.- The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of twenty-five (25) plug assemblies, 25 plugs, 25 cups, 25 pins, and 25 insulators which have been produced by the contractor or furnished by a supplier and which have been manufactured using the same production processes, procedures, and equipment which will be used in fulfilling the contract. All parts and materials shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and he shall submit a record of this inspection with the sample, including certificates of conformance of materials. A sample containing known defects shall not be sumitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

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4.2.2 Inspections to be performed.- Components will be subjected by the Government to any or all of the examinations specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.- If any component fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a component to comply with any of the stated requirements.

4.3 Inspection provisions

4.3.1 Lot formation.- The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded primers shall contain:

a. Parts from not more than one lot interfix number from one supplier.

b. Base charge from not more than one batch or blend of batches which results in homogeneity of all charge composition in the lot.

4.3.2 Examination.- The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major

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defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraphs of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cup (see dwg. 9225843 covering a detail of dwg. 9225836).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Split, cut, or crack.....	Visual	01001
Major:	AQL 0.40 percent		
101.	Thickness through bottom.....	Gage	01002
102.	Dent or distortion.....	Visual	01003
Minor:	AQL 1.50 percent		
201.	Outside diameter, min.....	Gage	01004
202.	Inside diameter maximim (max.)....	Gage	01005
203.	Evidence of poor workmanship other than critical above (see 3.5).....	Visual	01006

4.3.2.2 Plug (see dwg. 9225841 covering a detail of dwg. 9225836).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Diameter of hole, min.....	Gage	02001
102.	Depth of hole counterbore.....	Gage	02002

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Minor:	AQL 1.50 percent		
201.	Diameter of hole counter-bore, max.....	Gage	02003
202.	Chamfer missing.....	Visual	02004
203.	Evidence of poor workmanship (see 3.5).....	Visual	02005

4.3.2.3 Plug assembly (see dwg. 9225838 covering a detail of dwg. 9225836).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Length of plug.....	Gage	03001
102.	Length of pin.....	Gage	03002
103.	Concentricity of pin with plug diameter.....	Gage	03003
104.	Insulator protrusion on pin over max.....	Gage	03004
105.	Plating missing, contains bare spot, or excessively oxidized..	Visual	03005
106.	Evidence of poor workmanship (see 3.5).....	Visual	03006
Minor:	None defined		

4.3.2.4 Bridge and plug assembly prior to assembly of spot charge (see dwg. 9225837 covering a detail of dwg. 9225836).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Bridge missing or broken.....	Visual	04001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5).....	Visual	04002

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4.3.2.5 Bridge and plug assembly (see dwg. 9925837 covering a detail of dwg. 9225836).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Spot charge missing.....	Visual	05001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5).....	Visual	05002

4.3.2.6 Assembly (see dwg. 9225836).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Crack, split, or cut in cup.....	Visual	06001
Major:	AQL 1.00 percent		
101.	Total length.....	Gage	06002
102.	Diameter, max.....	Gage	06003
103.	Length of cup.....	Gage	06004
104.	Evidence of poor workmanship on contact surface of pin (see 3.5).	Visual	06005
105.	Joint not sealed.....	Visual	06006
106.	Crimp not full 360 degrees.....	Visual	06007
Minor:	AQL 1.00 percent		
201.	Deep grooves or scratches in cup.	Visual	06008
202.	Evidence of poor workmanship other than in critical above or on pin surface (see 3.5).....	Visual	06009

4.3.3 Testing.- Unless otherwise specified herein, the following tests shall be performed using Standard MIL-STD-105 for sampling plan requirements.

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4.3.3.1 Moisture content of base charge (see 3.2) - Major defect - Code No. 07001.- The contractor shall provide adequate controls to insure that the composition complies with the requirements. For verification, the contractor shall select a sample for test from that quantity used in each eight hours of loading production. A composite sample shall not be used. If the moisture content of a sample exceeds the requirements and loading has not begun, that quantity of composition represented by the sample shall be rejected. If components have been loaded with composition containing excessive moisture, the remaining unloaded composition and the loaded components shall be rejected. The test procedure shall be as specified in 4.4.1.

4.3.3.2 Functioning (see 3.3 and Table I).

4.3.3.2.1 Sampling.- One hundred and eighty-five (185) primer assemblies shall be selected from each lot for test. The number of samples to be subjected to each test shall be as specified in Table I.

TABLE I

Test	Classification	Number of Samples	Code No.
Time (3.3.1)	Major	35	06010
Sound power output (see 3.3.2)	Major	50 (see 4.3.3.6)	06011
Duds (see 3.3.3)	Major	100 *	06012

*Primers that dud in the Time test phase shall be recorded and replaced from this sample.

4.3.3.2.2 Rejection.- The lot shall be rejected if any of the following test results occur:

a. Time - The mean functioning time ($\bar{X}$) plus the product of 2.27 and the standard deviation (s) of the sample exceed 300 microseconds (see 6.4)

b. Sound power output - Two (2) or more primer assemblies fail to comply with the sound power output requirements. Duds shall not be considered as sound power output failures.

c. Duds - The total number of duds found during the time, sound power output and dud tests equals or exceeds six (6).

d. Total defects - The total number of defectives found in b, c, and the 4.3.3.6 resistance test equals or exceeds seven (7).

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4.3.3.3 Plug assembly push test (see dwg. 9225838) Major defect - Code No. 03007.- The sampling plan for this test shall be in accordance with code letter L with an AQL of 0.40 percent. The test shall be performed as specified in 4.4.3. Plug assemblies subjected to this test shall not be returned to the lot.

4.3.3.4 Resistance between pin and plug (see dwg. 9225838) - Major defect - Code No. 03008.- This test shall be performed 100 percent in accordance with the procedure specified in 4.4.4. Any plug assembly that fails to comply with the requirement shall be classed defective and removed from the lot.

4.3.3.5 Resistance between pin and cup (see dwg. 9225836) - Major defect - Code No. 06013.- This test shall be performed 100 percent in accordance with the procedure specified in 4.4.5. Any primer assembly that fails to comply with the requirement shall be classed defective and removed from the lot.

4.3.3.6 Resistance between pin and cup after water and Freon TF conditioning (see 3.4) - Major defect - Code No. 06014.- The fifty samples selected in Table I for the sound power output test shall be used for this test. If three (3) or more primer assemblies fail to comply with the resistance limits after conditioning, the lot shall be rejected. The test shall be performed as specified in 4.4.6.

4.3.3.7 Base charge composition (see dwg. 9225836) - Major defect - Code No. 08001.- The contractor shall select for test a sample of approximately five (5) grams from that quantity of base charge composition used in each eight hours of loading production. A composite sample shall not be used. If the sample fails to comply with the composition requirements, the quantity of composition represented by the sample shall be rejected, including any primer assemblies loaded with the non-conforming material. The test shall be performed as specified in 4.4.7.

4.3.4 Inspection equipment.- Equipment Tabulation Number ET-9225836 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

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4.3.4.1 Government rights to documentation.- Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.4 Test methods and procedures

4.4.1 Moisture content of base charge.- A weighed portion of approximately five (5) grams shall be transferred to a tared weighing bottle. The bottle shall be dried at one hundred (100) to one hundred and five (105) degrees Centigrade (C) for three (3) hours min, then cooled in a desiccator and weighed. Calculate the loss in weight as percent moisture in the sample.

4.4.2 Functioning

4.4.2.1 Time.- The primer asseblies shall be assembled in the test fixture specified in 4.3.4 and fired by applying the energy from a nineteen and eight tenths (19.8) max. microfarad capacitor charged to two and two tenths (2.2) volts max. The firing time shall be observed for compliance with the requirements.

4.4.2.2 Sound power output.- The primer assemblies shall be immersed for forty-eight (48) hours, min. in Freon TF maintained at a temperature of one hundred and forty-five (145) plus or minus five (5) degrees Fahrenheit (F.). Not sooner than two (2) hours after removed from the Freon TF, the samples shall be immersed for twenty-four (24) hours min. in water maintained at the above temperature. After conditioning, the primer assembly shall be assembled in the test equipment specified in 4.3.4 and fired by applying the energy specified in 4.4.2.1.

4.4.2.3 Duds.- The primer assembly shall be assembled in the applicable test equipment specified in 4.3.4 and fired by applying the energy specified in 4.4.2.1.

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4.4.3 Plug assembly push test.- The plug assembly shall be assembled in the test equipment specified in 4.3.4. The required axial force shall be applied to the pin in the direction indicated. The assembly shall be observed during application of the load to determine compliance with the requirement.

4.4.4 Resistance between pin and plug.- The plug assembly shall be connected in an electrical circuit containing a three hundred and twenty-five (325) volt direct current power source and a resistance measuring instrument accurate within three (3) percent for the full scale. The pin shall be used as one electrode and the plug as the other. The meter shall be observed for compliance with the requirement.

4.4.5 Resistance between pin and cup.- The primer assembly shall be connected to the test equipment specified in 4.3.4 and the resistance measured. The input current shall be limited to one (1) milliampere and a suitable barricade should be used. The meter shall be observed for compliance with the requirement.

4.4.6 Resistance between pin and cup after Freon TF and water immersion.- The primer assemblies shall be tested as specified in 4.4.5 after Freon TF conditioning and again after water conditioning. The meter shall be observed for compliance with the requirement after each test.

4.4.7 Determination of base charge composition.

4.4.7.1 Aluminum.- Transfer an accurately weighed portion of approximately two (2) grams of the sample to a tared medium porosity sintered glass crucible. Extract the sample with ten (10) twenty-five (25) milliliter (ml.) portions of distilled water at a temperature of 0 to 5 degrees Centigrade (C.) a filtering flask and suction. Stir the contents of the crucible with each portion of water for one (1) minute before applying suction. Transfer the extract quantitatively to a four hundred (400) ml. beaker using approximately fifty (50) ml. of wash water and reserve the extract and washings for the barium nitrate determinations. Rinse the undissolved residue with alcohol and ether and aspirate until no further solvent odors are apparent. Place the crucible and contents in a drying oven at 100 to 105 degrees C. for one (1) hour, cool in a desiccator and weigh. Calculate the increase in weight of the crucible to percent of aluminum powder on a moisture free basis as follows:

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$$\text{Percentage of aluminum powder} = \frac{100\,A}{W\,(1-M)}$$

where: A = increase in weight of filtering crucible
W = weight of sample
M = percent moisture expressed as a decimal fraction

4.4.7.2 Barium nitrate.- To the extract reserved during the aluminum powder determination, add one ml. of 12N hydrochloric acid. Heat to boiling on a hot plate and slowly add 50 ml. of 3N (1:8) sulferic acid from a 50 ml. pipette. Allow the mixture to boil gently for five (5) minutes and finally to digest for at least two (2) hours on a steam bath. Collect the precipitated barium sulfate in a tared Gooch crucible which has been previously heated in a muffle furnace at approximately 750 degrees C. for 30 minutes. Wash the precipitate ten (10) times with 25 ml. portions of hot water, dry in an oven at 100 plus or minus 5 degrees C. for one hour and then in a muffle furnace at approximately 750 degrees C. for 30 minutes, cool in a desiccator and weigh. Calculate the percentage of barrium nitrate in the sample on a moisture free basis as follows:

$$\text{Percentage of barium nitrate} = \frac{111.98A}{W\,(1-M)}$$

where: A = increase in weight of Gooch crucible
W = weight of sample
M = percent moisture expressed as a decimal fraction

4.4.7.3 Potassium perchlorate.- Calculate the percentage of potassium perchlorate in the sample by deducting the percentages of aluminum powder and barium nitrate from 100.

5. PREPARATION FOR DELIVERY

5.1 Packaging

5.1.1 Level C.- The eleectric primers shall be packaged in accordance with Code of Federal Regulations 49 CFR Parts 171-179.

5.2 Packing

5.2.1 Level C.- The electric primers shall be packed in accordance with Code of Federal Regulations 49 CFR Parts 171-179 and in a manner that will insure safe delivery from the point

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of manufacture to the first receiving activity.

5.3 Marking.- Each unit container shall be marked in accordance with standard commercial marking requirements. Each exterior container shall be marked in accordance with Drawing 8796522. The Dot marking shall be "(Quantity) ELECTRIC BLASTING CAPS - HANDLE CAREFULLY".

6. NOTES

6.1 Ordering data.- Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.- Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Provisions for submission of first article samples.

6.2 Inspection code numbers.- The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. In addition, they should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.5).

6.3 Inspection equipment.- The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).- Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.2 Inspection equipment list codes.- The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.- Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.- Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category A, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.5 Submission of contractor designs.- All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND15. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Standard deviation.- The following formula or its equivalent, shall be used to compute the sample standard deviation:

$$s = \sqrt{\frac{\Sigma(\bar{X} - X)^2}{n-1}}$$

where: s = standard deviation
X = numerical value of individual time observation
n = sample size
$\bar{X}$ = average value of the sample

$\Sigma(\bar{X} - X)^2$ = the sum of the square of the sample mean time minus each individual time.

6.5 Check test for possible deterioration.- The following test is designed to prevent loading of primers which have deteriorated into the ammunition component with which it is to be loaded. The tests should be performed on primers from each lot which have been stored for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the primers are loaded does not require that the check tests be performed, the contracting officer should insert provisions for the performance of the check tests in the contract. The contracting officer should arrange to have

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around the battery cap and firing plug, or

the contractor reimbursed for any expense incurred in the performance of the check test for possible deterioration. Two hundred (200) primer assemblies shall be selected from each lot and tested as follows:

6.5.1 Functioning time and duds.- Forty (40) primers shall be subjected to the test specified in 4.4.2.1. Observation shall be made for the functioning time and all duds that may occur shall be recorded. Duds should be replaced by primers from the 200 selected. If the mean time ($\overline{X}$) plus 2.03 times the standard deviation (s) exceeds three hundred (300) microseconds, the lot shall be rejected. The standard deviation shall be computed from the formula in 6.4 or its equivalent. Duds shall not enter into the calculation for standard deviation.

6.5.2 Sound power output and duds.- Fifty (50) primer assemblies shall be selected and subjected to the test specified in 4.4.2.2. The sound power output and the number of duds that occur shall be recorded. Duds shall be replaced from the 200 selected. If three (3) or more primer assemblies fail to comply with the sound power output requirements, the lot shall be rejected.

6.5.3 Duds.- The remainder of the primer assemblies selected in 6.5 shall be tested as specified in 4.4.2.3. The lot shall be rejected if the total number of duds accumulated in 6.5.1, 6.5.2, and 6.5.3 equals or exceeds eight (8).

Custodian:
Army-MU

Preparing Activity
Army-MU

Project Number: 1375-A-035

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Check the source to verify that this is the currentversion before use.